

Section 2 Answers

-- Table creation

```
CREATE TABLE Department (  
    DeptID INT PRIMARY KEY,  
    DeptName VARCHAR(100) NOT NULL,  
    Budget DECIMAL(18, 2)  
);  
  
CREATE TABLE Employee (  
    EmpID INT PRIMARY KEY,  
    EmpName VARCHAR(100) NOT NULL,  
    DeptID INT,  
    Salary DECIMAL(18, 2),  
    CONSTRAINT FK_Employee_Department FOREIGN KEY (DeptID)  
        REFERENCES Department(DeptID)  
);  
  
CREATE TABLE Project (  
    ProjectID INT PRIMARY KEY,  
    ProjectName VARCHAR(100) NOT NULL,  
    DeptID INT,  
    AllocatedFunds DECIMAL(18, 2),  
    CONSTRAINT FK_Project_Department FOREIGN KEY (DeptID)  
        REFERENCES Department(DeptID)  
);  
  
--Value insertion  
INSERT INTO Department (DeptID, DeptName, Budget) VALUES  
(101, 'Human Resources', 500000.00),  
(102, 'Engineering', 1200000.00),  
(103, 'Marketing', 350000.00);  
  
INSERT INTO Employee (EmpID, EmpName, DeptID, Salary) VALUES  
(1, 'Alice Johnson', 102, 85000.00),  
(2, 'Bob Smith', 101, 60000.00),  
(3, 'Charlie Brown', 102, 92000.00),  
(4, 'Diana Prince', 103, 75000.00),  
(5, 'Edward Norton', 102, 88000.00);  
  
INSERT INTO Project (ProjectID, ProjectName, DeptID, AllocatedFunds) VALUES  
(501, 'Cloud Migration', 102, 200000.00),  
(502, 'Annual Recruitment', 101, 50000.00),  
(503, 'Summer Campaign', 103, 120000.00),  
(504, 'AI Research', 102, 450000.00);  
  
--Q1  
  
GO  
CREATE PROCEDURE GetDepartmentInfo  
    @DeptID INT  
AS  
BEGIN  
    SELECT DeptID, DeptName, Budget  
    FROM Department  
    WHERE DeptID = @DeptID;  
END;  
EXEC GetDepartmentInfo @DeptID = 101;
```

	DeptID	DeptName	Budget
1	101	Human Resources	500000.00

```
--Q2
GO
CREATE PROCEDURE GetEmployeeSalary
    @EmpID INT,
    @Salary DECIMAL(18, 2) OUTPUT
AS
BEGIN
    SELECT @Salary = Salary
    FROM Employee
    WHERE EmpID = @EmpID;
END;
DECLARE @MyFolder DECIMAL(18, 2);

EXEC GetEmployeeSalary
    @EmpID = 1,
    @Salary = @MyFolder OUTPUT;

SELECT @MyFolder AS 'The Salary We Found';
```

Commands completed successfully.

Completion time: 2026-04-06T13:52:09.0015097+05:30

```
--Q3
GO
CREATE PROCEDURE UpdateEmployeeSalary
    @EmpID INT,
    @NewSalary DECIMAL(18, 2)
AS
BEGIN
    IF @NewSalary > 0
    BEGIN
        UPDATE Employee
        SET Salary = @NewSalary
        WHERE EmpID = @EmpID;
    END
    ELSE
    BEGIN
        PRINT 'Update terminated: Salary cannot be zero or negative.';
    END
END;
EXEC UpdateEmployeeSalary @EmpID = 1, @NewSalary = 95000.00;

SELECT * FROM Employee WHERE EmpID = 1;
```

Results		Messages		
	EmpID	EmpName	DeptID	Salary
1	1	Alice Johnson	102	95000.00

Negative

```
EXEC UpdateEmployeeSalary @EmpID = 1, @NewSalary = -500.00;
```

Messages	
Update terminated: Salary cannot be zero or negative.	
Completion time: 2026-04-06T14:00:58.8583658+05:30	

--Q4

GO

```
CREATE PROCEDURE CountEmployeesInDept
    @DeptID INT
```

AS

BEGIN

```
    DECLARE @EmployeeCount INT;
```

```
    SELECT @EmployeeCount = COUNT(*)
    FROM Employee
    WHERE DeptID = @DeptID;
```

```
    PRINT 'Total employees in this department: ' + CAST(@EmployeeCount AS
    VARCHAR(10));
```

END;

```
EXEC CountEmployeesInDept @DeptID = 102;
```

Messages	
Total employees in this department: 3	
Completion time: 2026-04-06T14:04:13.6846516+05:30	

--Q5

GO

```
CREATE PROCEDURE InsertEmployee
    @EmpID INT,
    @EmpName VARCHAR(100),
    @DeptID INT,
    @Salary DECIMAL(18, 2)
```

AS

BEGIN

```
    BEGIN TRY
```

```

        INSERT INTO Employee (EmpID, EmpName, DeptID, Salary)
        VALUES (@EmpID, @EmpName, @DeptID, @Salary);
    END TRY
    BEGIN CATCH
        PRINT 'Error creating employee record! ' + ERROR_MESSAGE();
    END CATCH
END;

```

```

EXEC InsertEmployee
    @EmpID = 99,
    @EmpName = 'John Doe',
    @DeptID = 101,
    @Salary = 55000.00

```

```

SELECT * FROM Employee WHERE EmpID = 99;

```

	EmpID	EmpName	DeptID	Salary
1	99	John Doe	101	55000.00

Error message

```

EXEC InsertEmployee
    @EmpID = 99,
    @EmpName = ' John',
    @DeptID = 101,
    @Salary = 55000.00;

```

```

record! Violation of PRIMARY KEY constraint 'PK_Employee_AF2DBA79ACFD9FE6'. Cannot insert duplicate key in object 'dbo.Employee'. The duplicate key value is (99).
--06T14:12:58.5864625+05:30

```

--Q6

GO

```

CREATE VIEW Public_Employee_Directory AS
SELECT

```

```

    EmpID,
    EmpName,
    DeptID
FROM Employee;
GO

```

```

SELECT * FROM Public_Employee_Directory;

```

	EmpID	EmpName	DeptID
1	1	Alice Johnson	102
2	2	Bob Smith	101
3	3	Charlie Brown	102
4	4	Diana Prince	103
5	5	Edward Norton	102
6	99	John Doe	101